

LAPORAN PRAKTIKUM

TEKNOLOGI IOT II

*Komunikasi Asinkron dan WebSockets, Topologi Jaringan Terdesentralisasi
(ESP Now dan Mesh)*



Disusun Oleh:

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TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

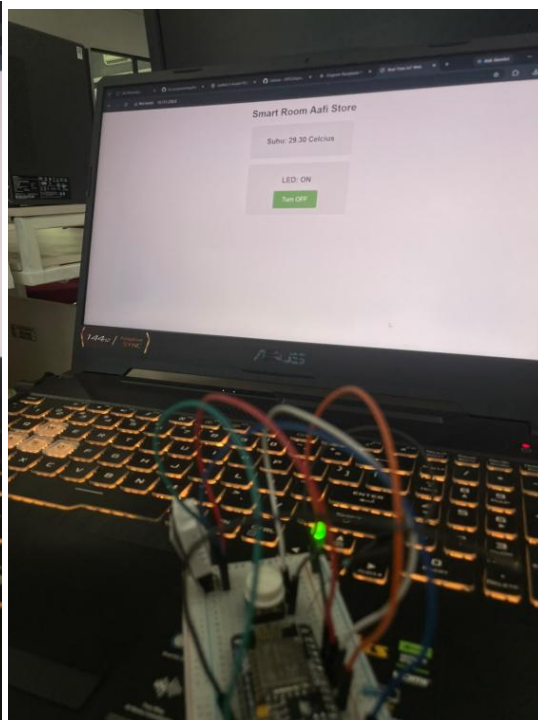
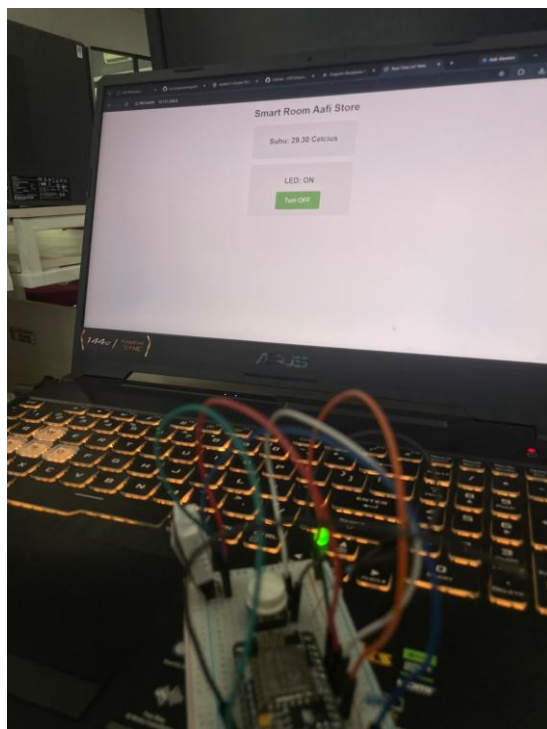
POLITEKNIK NEGERI LAMPUNG

2026/2027

Modul 4: Komunikasi Asinkron dan WebSockets

4.5.1 Praktikum 1: *Real-Time Dashboard* & Sinkronisasi Tombol Fisik

```
4.5.1Praktikum1ino
1 #include <ESP8266WiFi.h>
2 #include <ESPAsyncTCP.h>
3 #include <ESPAsyncWebServer.h>
4 #include <DHT.h>
5
6 const char* ssid = "V2030";
7 const char* password = "raid_0507";
8
9 // Konfigurasi Pin
10 const byte dhtPin = 2; // D4 (GPIO 2)
11 const byte buttonPin = 4; // D2 (GPIO 4)
12 const byte ledPin = 12; // D6 (GPIO 12)
13
14 DHT dht(dhtPin, DHT22);
15
16 // Variabel Pelacak Status (State & Cache)
17 bool ledState = false;
18 String currentTemp = "...";
19 String currentHum = "...";
20
21 int buttonState;
22 int lastButtonState = LOW;
23 unsigned long lastDebounceTime = 0;
```



Latihan 1 (Teori Jaringan): Jelaskan dengan kata-kata Anda sendiri, mengapa teknologi koneksi *WebSockets* jauh lebih hemat konsumsi data/bandwidth internet dan tidak membuat layar berkedip jika dibandingkan dengan metode *Meta Refresh HTTP* HTML yang Anda gunakan pada Tugas Praktikum Modul 3?

Penggunaan *WebSockets* jauh lebih hemat bandwidth dibandingkan metode *Meta Refresh HTTP* karena *WebSockets* hanya melakukan satu kali proses jabat tangan

(handshake) di awal untuk membuka terowongan komunikasi dua arah yang persisten. Setelah terowongan ini terbuka, NodeMCU dan browser hanya mentransfer paket data berukuran sangat kecil dalam format JSON tanpa perlu mengirimkan ulang seluruh dokumen HTML, CSS, dan file skrip secara berulang-ulang.

Latihan 2 (Analisis Waktu/Timer): Apa bahaya sistematis yang akan terjadi pada komunikasi *WebSockets* (reaksi antara *browser* dan NodeMCU) jika Anda bersikeras nekat memanggil sintaks `delay(3000)` di dalam siklus fungsi `void loop()` pada draf arsitektur kompleks ini?

Jika sintaks `delay(3000)` dipanggil di dalam siklus `void loop()`, dampak buruk utama yang akan terjadi adalah pembekuan eksekusi CPU NodeMCU secara total selama 3 detik pada setiap putaran program.

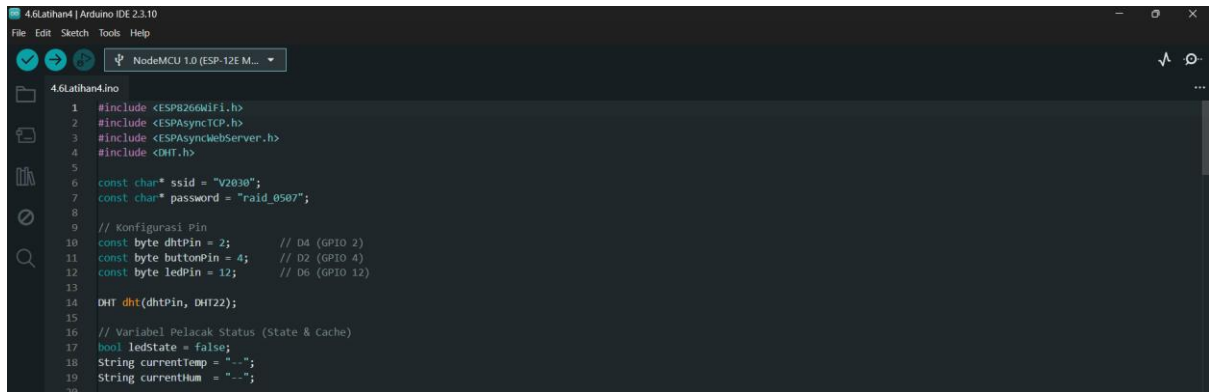
Latihan 3 (Modifikasi Front-End): Analisislah fungsi JavaScript pada blok // Merombak UI Tombol & Teks sesuai status hardware. Nama ID elemen HTML (`<span id="...">`) apa saja yang menjadi target injeksi/pengubahan teks oleh *script* yang ditulis oleh Anda?

Di bagian skrip JavaScript itu, ada tiga ID elemen HTML yang diubah-ubah teksnya. Pertama ada ID `tempValue` di tag `<span>` buat nampilin angka suhu dari sensor DHT. Kedua ada ID `ledStatus` di tag `<span>` buat ngubah status LED jadi tulisan ON atau OFF. Terakhir ada ID `toggleBtn` di tag `<button>` yang teksnya diganti jadi Turn ON atau Turn OFF, sekalian diganti class CSS-nya biar warna tombolnya berubah dari hijau ke merah atau sebaliknya.

Latihan 4 (Pengembangan Telemetry Sensor - Kelembapan): Pada Praktikum 1, antarmuka web hanya menyajikan data suhu (*temperature*). Modifikasilah program agar sistem dapat membaca dan menampilkan data kelembapan (*humidity*) secara *real-time* via *WebSockets*:

- i. **Sisi Back-End (C++):** Buat variabel penampung (misal: `String currentHum = "--";`). Baca nilai kelembapan menggunakan fungsi `dht.readHumidity()` di dalam siklus timer *non-blocking* `millis()`. Perbarui fungsi `notifyClients()` agar menyisipkan data kelembapan ke dalam paket JSON yang dikirim ke *browser* (contoh format: `{"led": "...", "suhu": "...", "hum": "..."}.`
- ii. **Sisi Front-End (HTML & JavaScript):** Tambahkan elemen kartu (`<div class="card">`) baru pada kerangka HTML sebagai penampil kelembapan (misal: `<h2>Kelembapan: <span id="humValue">--</span> %</h2>`). Kemudian, modifikasi fungsi JavaScript `onMessage(event)` agar mengekstrak data

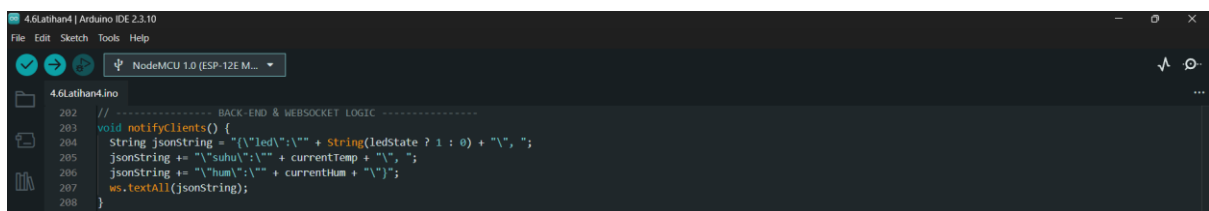
kelembapan dari objek JSON dan memperbarui teks pada elemen humValue.
Tuliskan baris kode modifikasi Anda (pada sisi HTML, JavaScript, dan C++)!



```
4.6Latihan4 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...)

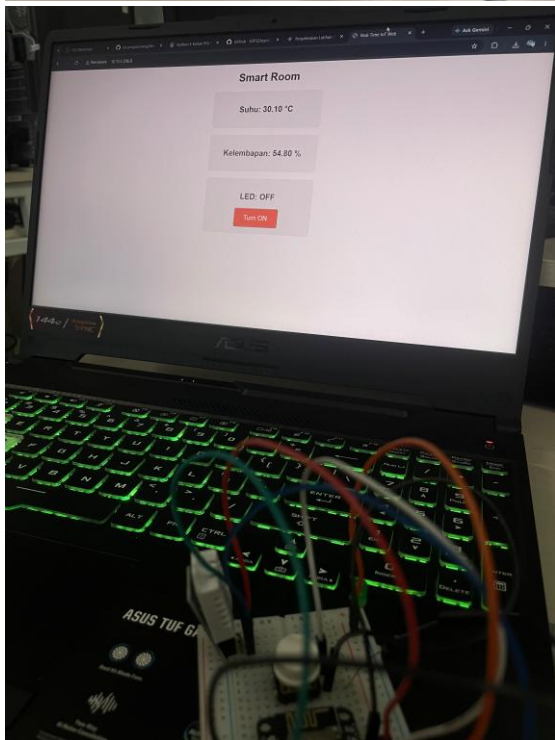
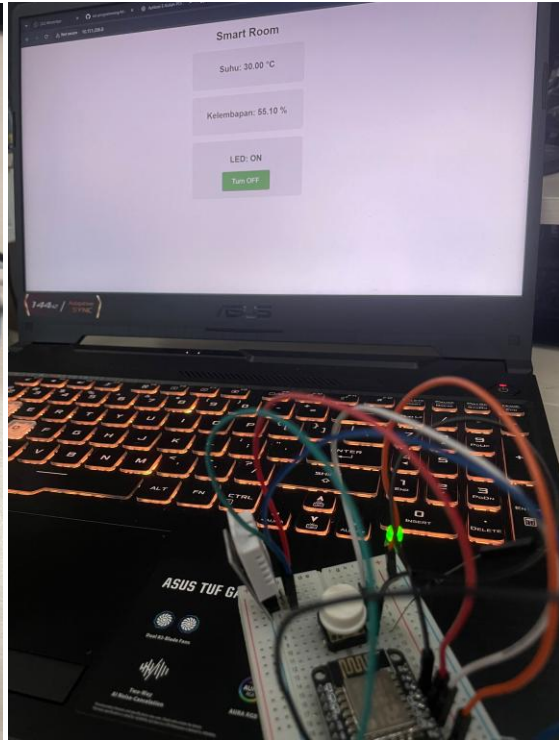
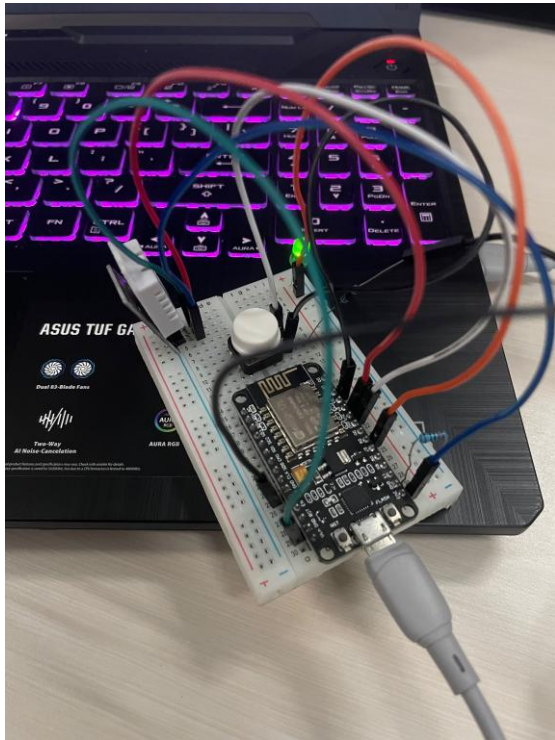
4.6Latihan4.ino
1 #include <ESP8266WiFi.h>
2 #include <ESPAsyncTCP.h>
3 #include <ESPAsyncWebServer.h>
4 #include <DHT.h>
5
6 const char* ssid = "V2030";
7 const char* password = "raid_0507";
8
9 // Konfigurasi Pin
10 const byte dhtPin = 2; // D4 (GPIO 2)
11 const byte buttonPin = 4; // D2 (GPIO 4)
12 const byte ledPin = 12; // D6 (GPIO 12)
13
14 DHT dht(dhtPin, DHT22);
15
16 // Variabel Pelacak Status (State & Cache)
17 bool ledState = false;
18 String currentTemp = "--";
19 String currentHum = "--";
20
```



```
4.6Latihan4 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...)

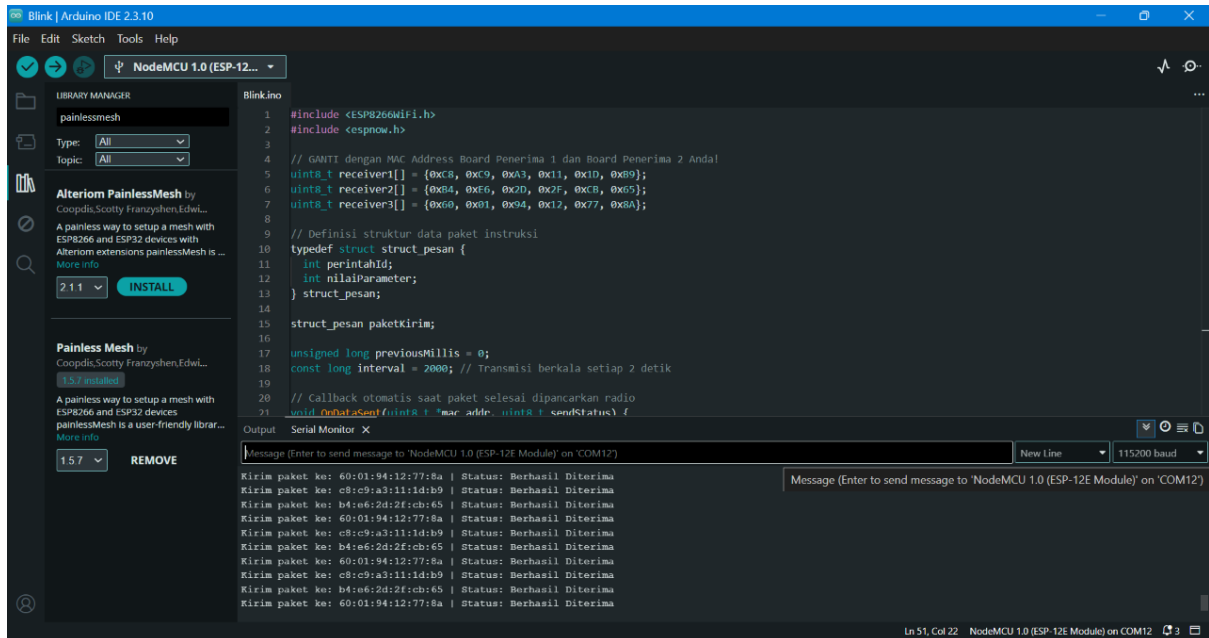
4.6Latihan4.ino
202 // ----- BACK-END & WEBSOCKET LOGIC -----
203 void notifyClients() {
204   String jsonString = "{\"led\": \"\" + String(ledState ? 1 : 0) + \"\", \"";
205   jsonString += "\"temp\": \"\" + currentTemp + \"\", \"";
206   jsonString += "\"hum\": \"\" + currentHum + \"\"}";
207   ws.textAll(jsonString);
208 }
```



Modul 5: Topologi Jaringan Terdesentralisasi (ESP-NOW dan Mesh)

5.5.2 Praktikum 2: Komunikasi Nirkabel One-to-Many (ESP-NOW)

Controller



```
1 #include <ESP8266WiFi.h>
2 #include <ESPNow.h>
3
4 // GANTI dengan MAC Address Board Penerima 1 dan Board Penerima 2 Andai
5 uint8_t receiver1[] = {0xC8, 0xE9, 0xA3, 0x1D, 0x80};
6 uint8_t receiver2[] = {0xB4, 0xE6, 0x2D, 0x2F, 0xC8, 0x55};
7 uint8_t receiver3[] = {0x60, 0x94, 0x12, 0x77, 0x8A};
8
9 // Definisi struktur data paket instruksi
10 typedef struct struct_pesanan {
11     int perintahId;
12     int nilaiParameter;
13 } struct_pesanan;
14
15 struct_pesanan paketKirim;
16
17 unsigned long previousMillis = 0;
18 const long interval = 2000; // Transmisi berkala setiap 2 detik
19
20 // Callback otomatis saat paket selesai dipancarkan radio
21 void OnDataSent(uint8_t *mac_addr, uint8_t sendStatus) {
```

Serial Monitor X

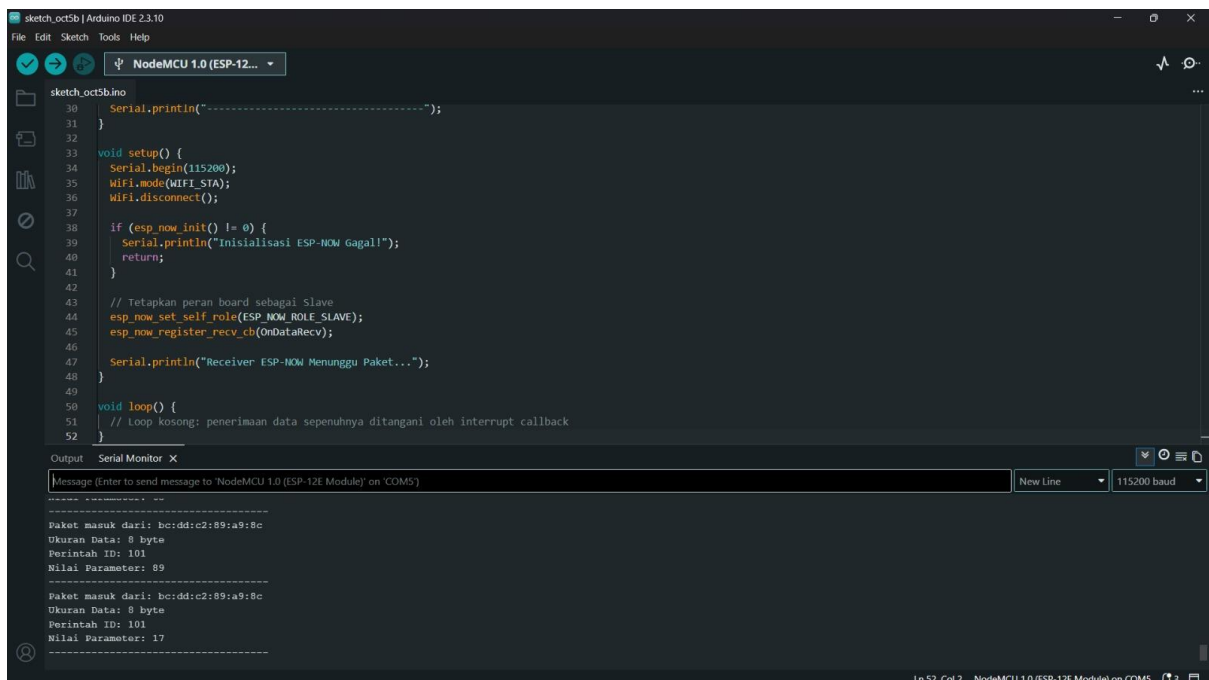
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')

115200 baud

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')

Kirim paket ke: 60:01:94:12:77:8A | Status: Berhasil Diterima
Kirim paket ke: c8:e9:a3:1d:b9 | Status: Berhasil Diterima
Kirim paket ke: b4:e6:2d:2f:cb:65 | Status: Berhasil Diterima
Kirim paket ke: 60:01:94:12:77:8A | Status: Berhasil Diterima
Kirim paket ke: c8:e9:a3:1d:b9 | Status: Berhasil Diterima
Kirim paket ke: b4:e6:2d:2f:cb:65 | Status: Berhasil Diterima
Kirim paket ke: 60:01:94:12:77:8A | Status: Berhasil Diterima
Kirim paket ke: c8:e9:a3:1d:b9 | Status: Berhasil Diterima
Kirim paket ke: b4:e6:2d:2f:cb:65 | Status: Berhasil Diterima
Kirim paket ke: 60:01:94:12:77:8A | Status: Berhasil Diterima

Penerima 1



```
30 Serial.println("-----");
31 }
32
33 void setup() {
34     Serial.begin(115200);
35     WiFi.mode(WIFI_STA);
36     WiFi.disconnect();
37
38     if (esp_now_init() != 0) {
39         Serial.println("Inisialisasi ESP-NOW Gagal!");
40         return;
41     }
42
43     // Tetapkan peran board sebagai Slave
44     esp_now_set_self_role(ESP_NOW_ROLE_SLAVE);
45     esp_now_register_recv_cb(OnDataRecv);
46
47     Serial.println("Receiver: ESP-NOW Menunggu Paket...");
48 }
49
50 void loop() {
51     // Loop kosong: penerimaan data sepenuhnya ditangani oleh interrupt callback
52 }
```

Serial Monitor X

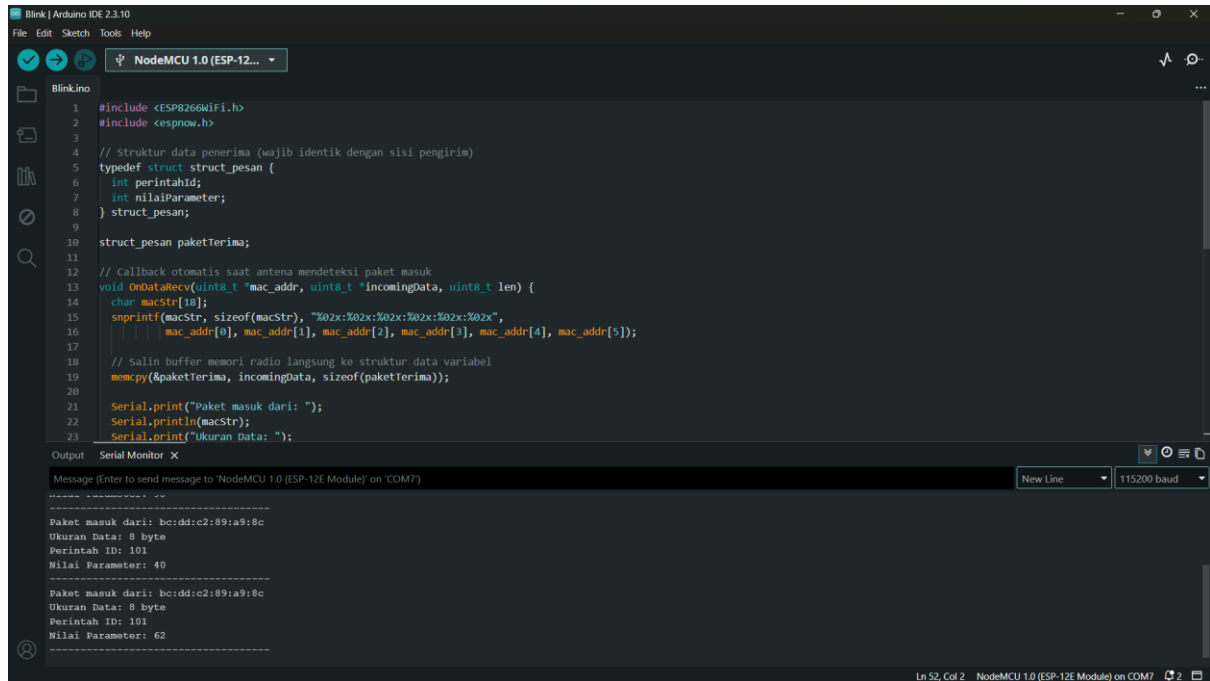
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

115200 baud

Paket masuk dari: bc:dd:c2:89:a9:8c
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 89

Paket masuk dari: bc:dd:c2:89:a9:8c
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 17

Penerima 2



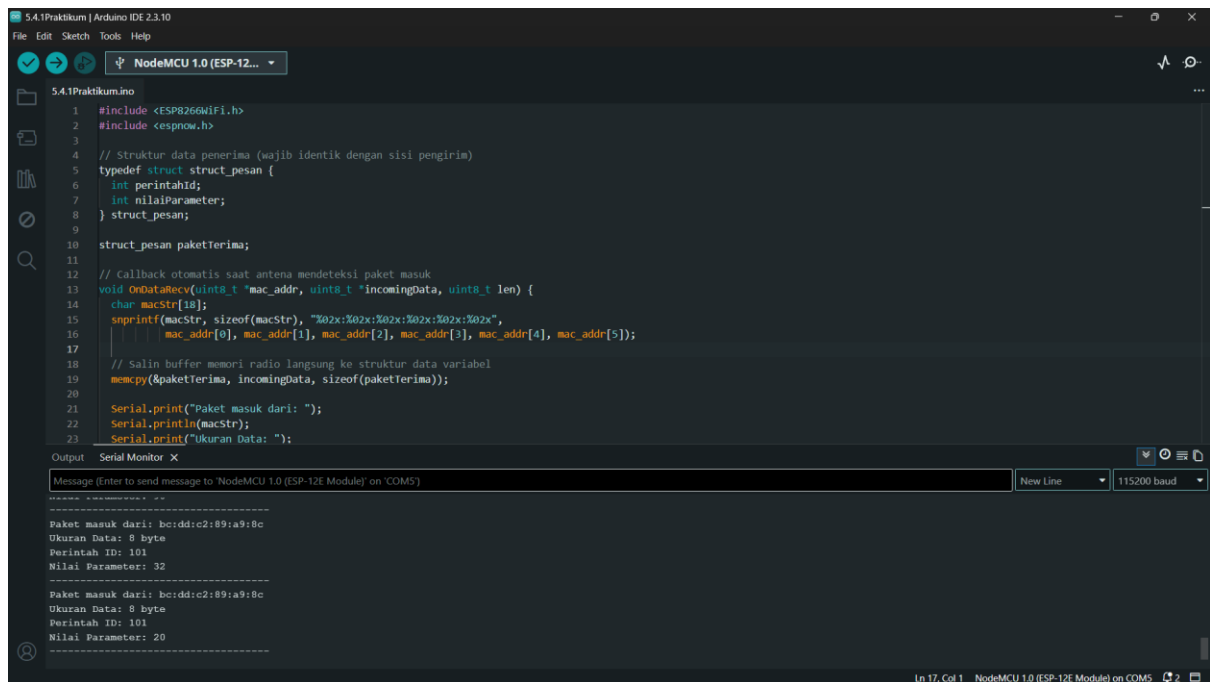
The screenshot shows the Arduino IDE interface with the file 'Blinkino' open. The code defines a structure for received packets and a callback function to process incoming data. The serial monitor shows two received packets with their respective MAC addresses, lengths, command IDs, and parameter values.

```
1 #include <ESP8266WiFi.h>
2 #include <espnow.h>
3
4 // Struktur data penerima (wajib identik dengan sisi pengirim)
5 typedef struct struct_pesan {
6     int perintahId;
7     int nilaiParameter;
8 } struct_pesan;
9
10 struct_pesan paketTerima;
11
12 // Callback otomatis saat antenna mendeteksi paket masuk
13 void OnDataRecv(uint8_t *mac_addr, uint8_t *incomingData, uint8_t len) {
14     char macStr[10];
15     sprintf(macStr, sizeof(macStr), "%02x:%02x:%02x:%02x:%02x",
16             mac_addr[0], mac_addr[1], mac_addr[2], mac_addr[3], mac_addr[4], mac_addr[5]);
17
18     // Salin buffer memori radio langsung ke struktur data variabel
19     memcpy(&paketTerima, incomingData, sizeof(paketTerima));
20
21     Serial.print("Paket masuk dari: ");
22     Serial.println(macStr);
23     Serial.print("ukuran Data: ");
```

Serial Monitor Output:

```
-----
Paket masuk dari: bc:dd:c2:89:a9:8c
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 40
-----
Paket masuk dari: bc:dd:c2:89:a9:8c
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 62
-----
```

Penerima 3



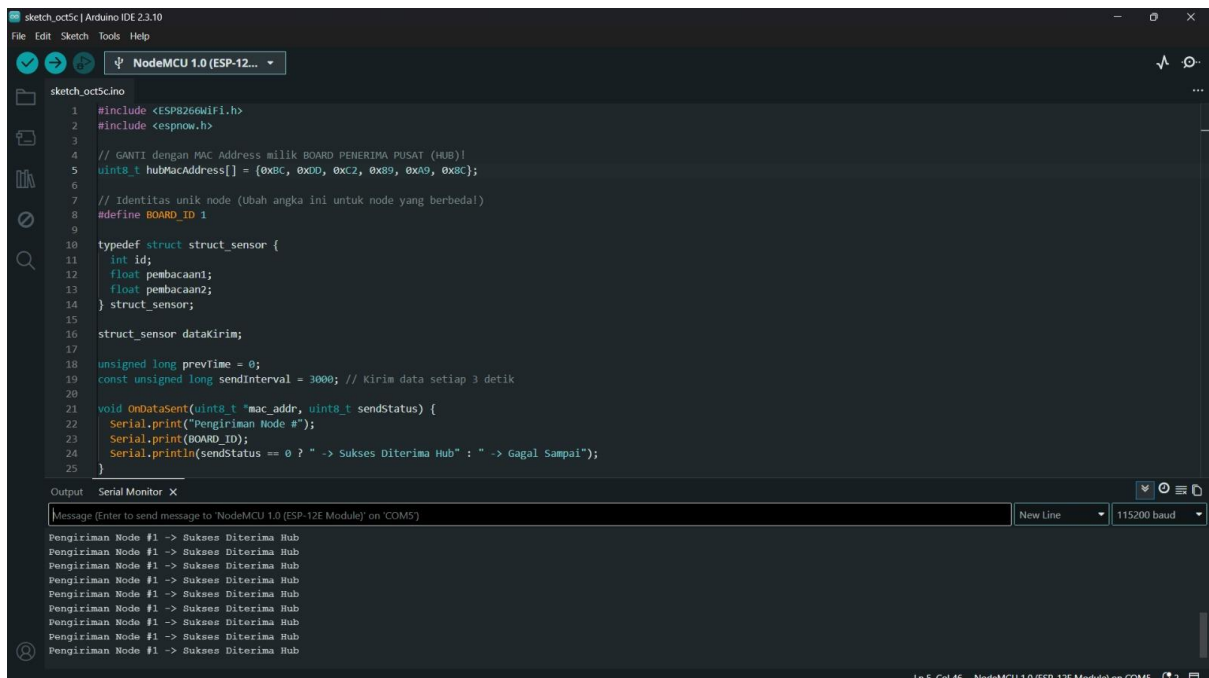
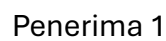
The screenshot shows the Arduino IDE interface with the file '5.4.1Praktikum.ino' open. The code is identical to the one in the previous image. The serial monitor shows two received packets with their respective MAC addresses, lengths, command IDs, and parameter values.

```
1 #include <ESP8266WiFi.h>
2 #include <espnow.h>
3
4 // Struktur data penerima (wajib identik dengan sisi pengirim)
5 typedef struct struct_pesan {
6     int perintahId;
7     int nilaiParameter;
8 } struct_pesan;
9
10 struct_pesan paketTerima;
11
12 // Callback otomatis saat antenna mendeteksi paket masuk
13 void OnDataRecv(uint8_t *mac_addr, uint8_t *incomingData, uint8_t len) {
14     char macStr[10];
15     sprintf(macStr, sizeof(macStr), "%02x:%02x:%02x:%02x:%02x",
16             mac_addr[0], mac_addr[1], mac_addr[2], mac_addr[3], mac_addr[4], mac_addr[5]);
17
18     // Salin buffer memori radio langsung ke struktur data variabel
19     memcpy(&paketTerima, incomingData, sizeof(paketTerima));
20
21     Serial.print("Paket masuk dari: ");
22     Serial.println(macStr);
23     Serial.print("ukuran Data: ");
```

Serial Monitor Output:

```
-----
Paket masuk dari: bc:dd:c2:89:a9:8c
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 32
-----
Paket masuk dari: bc:dd:c2:89:a9:8c
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 20
-----
```

Controller



Penerima 2

The image shows the Arduino IDE 2.3.10 interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. The toolbar contains icons for opening files, saving, and running. The main editor window displays a sketch named "NodeMCU 1.0 (ESP-12E...)" with the following code:

```
1 #include <ESP8266WiFi.h>
2 #include <espnow.h>
3
4 // GANTI dengan MAC Address milik BOARD PENERIMA PUSAT (HUB)!
5 uint8_t hubMacAddress[] = {0xB0, 0x00, 0xC2, 0xB9, 0xA9, 0xB0};
6
7 // Identitas unik node (ubah angka ini untuk node yang berbeda!)
8 #define BOARD_ID 2
9
10 typedef struct struct_sensor {
11     int id;
12     float pembacaan1;
13     float pembacaan2;
14 } struct_sensor;
15
16 struct_sensor dataKirim;
17
18 unsigned long prevTime = 0;
19 const unsigned long sendInterval = 3000; // Kirim data setiap 3 detik
20
21 void OnDataSent(uint8_t *mac_addr, uint8_t sendStatus) {
22     Serial.print("Pengiriman Node #");
23     Serial.print(BOARD_ID);
```

The bottom panel shows the "Serial Monitor" window, which is set to "New Line" and "115200 baud". It displays the following output:

```
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
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Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
```

The status bar at the bottom indicates "Ln 5, Col 45" and "NodeMCU 1.0 (ESP-12E Module) on COM7".

Penerima 3

The screenshot displays the Arduino IDE interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. The toolbar shows icons for opening files, saving, and running the sketch. The main editor window shows the code for `5.4.2Praktikum.ino`, which is a C++ program for an ESP8266 module. The code includes headers for `ESP8266WiFi.h` and `espnow.h`, defines a MAC address and a board ID, and implements a function `OnDataSent` to send data via ESP-NOW. The Serial Monitor is open at the bottom, showing the output of the program, which consists of multiple lines of the message "Pengiriman Node #3 -> Sukses Diterima Hub".

```
#include <ESP8266WiFi.h>
#include <espnow.h>

// GANTI dengan MAC Address milik BOARD PENERIMA PUSAT (HUB!)
uint8_t hubMacAddress[] = {0xBC, 0x00, 0x2, 0x89, 0xA9, 0xBC};

// Identitas unik node (Ubah angka ini untuk node yang berbeda!)
#define BOARD_ID 3

typedef struct struct_sensor {
    int id;
    float pembacaan1;
    float pembacaan2;
} struct_sensor;

struct_sensor dataKirim;

unsigned long prevtime = 0;
const unsigned long sendInterval = 3000; // Kirim data setiap 3 detik

void OnDataSent(uint8_t *mac_addr, uint8_t sendStatus) {
    Serial.print("Pengiriman Node #");
    Serial.print(BOARD_ID);
}
```

Serial Monitor X

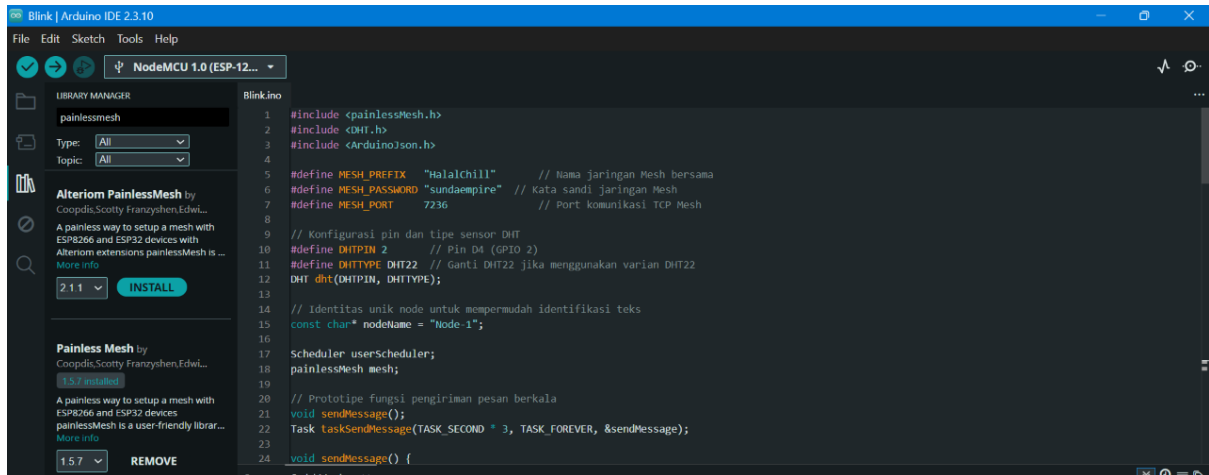
Message [Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5']

Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub
Pengiriman Node #3 -> Sukses Diterima Hub

Ln 5, Col 46 NodeMCU 1.0 (ESP-12E Module) on COM5

5.5.4 Praktikum 4: Jaringan Sensor Ad-Hoc Multi-Hop (painlessMesh)

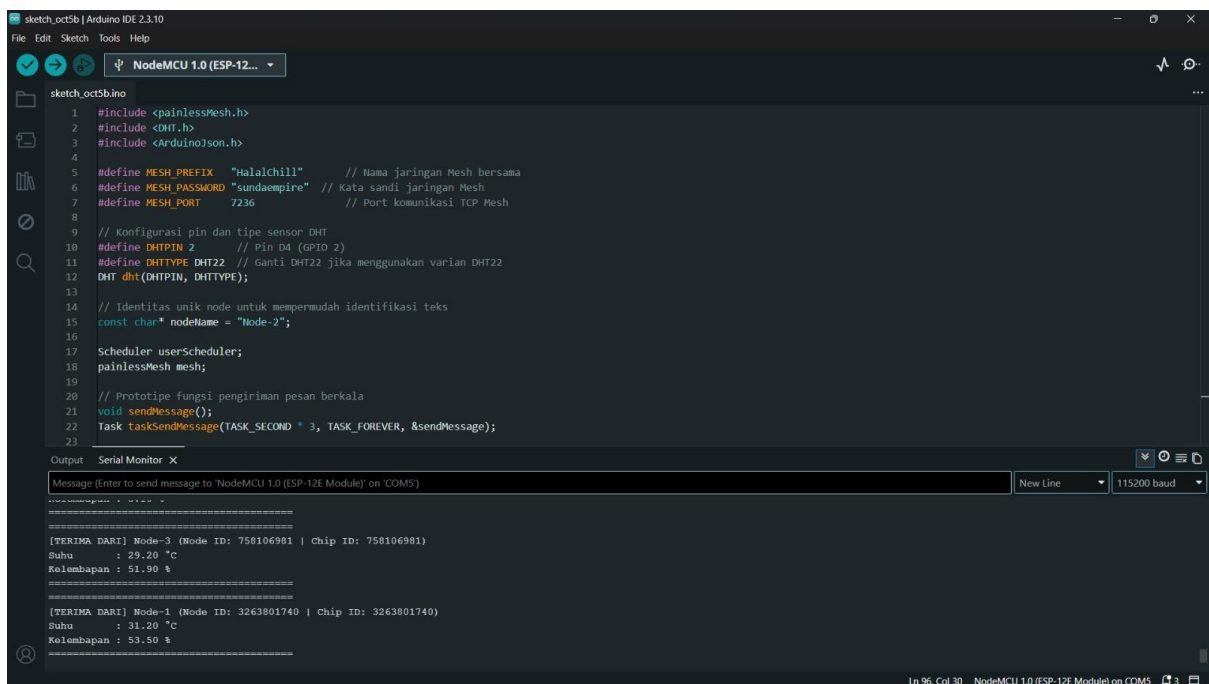
Controller



The screenshot shows the Arduino IDE interface with the 'Blink' sketch open. The 'LIBRARY MANAGER' on the left lists 'painlessMesh' by Coopdis, Scotty Franzyschen, Edwi... with version 2.1.1 installed. The main editor displays the code for 'Blink.ino'.

```
1 #include <painlessMesh.h>
2 #include <DHT.h>
3 #include <ArduinoJson.h>
4
5 #define MESH_PREFIX "Halalchilli" // Nama jaringan Mesh bersama
6 #define MESH_PASSWORD "sundaempire" // Kata sandi jaringan Mesh
7 #define MESH_PORT 7236 // Port komunikasi TCP Mesh
8
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-1";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototype fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
23
24 void sendMessage() {
```

Penerima 1



The screenshot shows the Arduino IDE interface with the 'sketch_oct5b' sketch open. The main editor displays the code for 'sketch_oct5b.ino'. The 'Output' window at the bottom shows the serial monitor output.

```
1 #include <painlessMesh.h>
2 #include <DHT.h>
3 #include <ArduinoJson.h>
4
5 #define MESH_PREFIX "Halalchilli" // Nama jaringan Mesh bersama
6 #define MESH_PASSWORD "sundaempire" // Kata sandi jaringan Mesh
7 #define MESH_PORT 7236 // Port komunikasi TCP Mesh
8
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-2";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototype fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
23
```

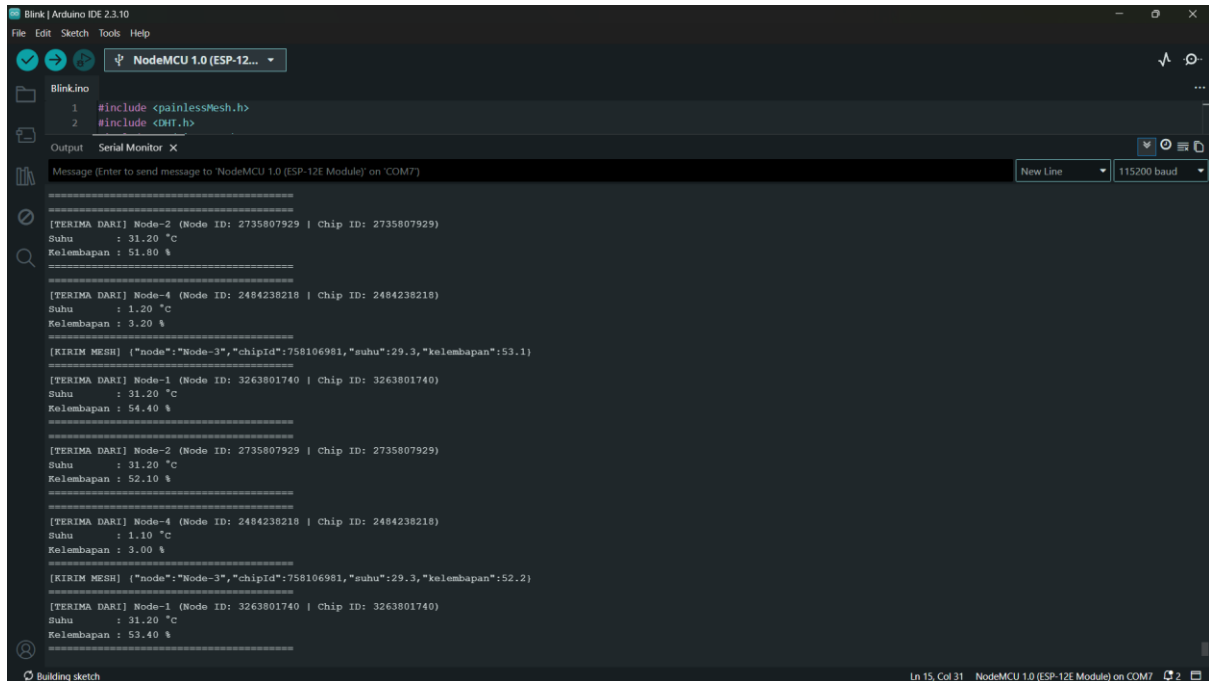
Output: Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

[TERIMA DARI] Node-3 (Node ID: 758106981 | Chip ID: 758106981)
Suhu : 29.20 °C
Kelembapan : 51.90 %

[TERIMA DARI] Node-1 (Node ID: 3263801740 | Chip ID: 3263801740)
Suhu : 31.20 °C
Kelembapan : 53.50 %

Penerima 2



```
1 #include <painlessMesh.h>
2 #include <DHT.h>

[TERIMA DARI] Node-2 (Node ID: 2735807929 | Chip ID: 2735807929)
Suhu : 31.20 °C
Kelembapan : 51.80 %

[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.20 °C
Kelembapan : 3.20 %

[KIRIM MESH] ("node":"Node-3","chipId":"758106981","suhu":29.3,"kelembapan":53.1)

[TERIMA DARI] Node-1 (Node ID: 3263801740 | Chip ID: 3263801740)
Suhu : 31.20 °C
Kelembapan : 54.40 %

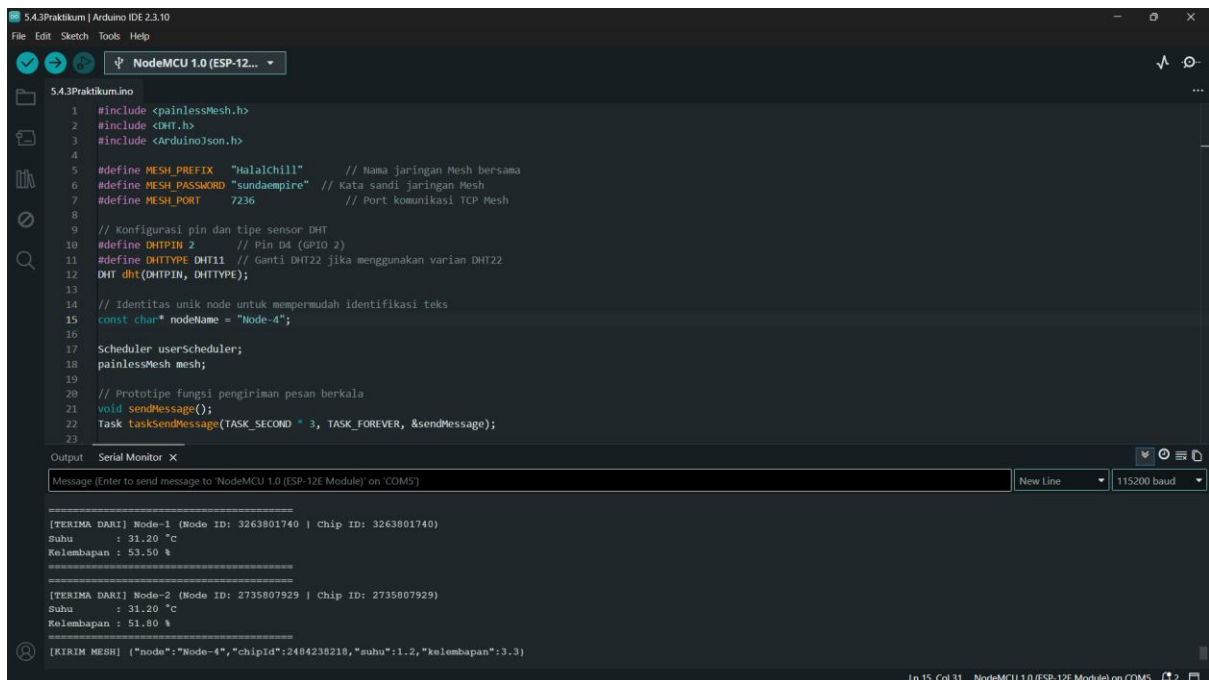
[TERIMA DARI] Node-2 (Node ID: 2735807929 | Chip ID: 2735807929)
Suhu : 31.20 °C
Kelembapan : 52.10 %

[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.10 °C
Kelembapan : 3.00 %

[KIRIM MESH] ("node":"Node-3","chipId":"758106981","suhu":29.3,"kelembapan":52.2)

[TERIMA DARI] Node-1 (Node ID: 3263801740 | Chip ID: 3263801740)
Suhu : 31.20 °C
Kelembapan : 53.40 %
```

Penerima 3



```
1 #include <painlessMesh.h>
2 #include <DHT.h>
3 #include <ArduinoJson.h>
4
5 #define MESH_PREFIX "Halalchill" // Nama jaringan Mesh bersama
6 #define MESH_PASSWORD "sundaempire" // Kata sandi jaringan Mesh
7 #define MESH_PORT 7236 // Port komunikasi TCP Mesh
8
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT11 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-4";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototipe fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
23
```

```
[TERIMA DARI] Node-1 (Node ID: 3263801740 | Chip ID: 3263801740)
Suhu : 31.20 °C
Kelembapan : 53.50 %

[TERIMA DARI] Node-2 (Node ID: 2735807929 | Chip ID: 2735807929)
Suhu : 31.20 °C
Kelembapan : 51.80 %

[KIRIM MESH] ("node":"Node-4","chipId":"2484238218","suhu":1.2,"kelembapan":3.3)
```

Latihan 1 (Eksperimen Deteksi ACK & Pemutusan Simpul - Praktikum 2): Saat program Praktikum 2 (ESP-NOW One-to-Many) sedang berjalan normal dan mencetak status "Berhasil Diterima" (sendStatus == 0)

ESP-NOW mendeteksi kegagalan pengiriman tanpa TCP melalui mekanisme MAC ACK frame di Layer 2 Wi-Fi (IEEE 802.11). Saat data dikirim secara *unicast*, chip radio penerima otomatis membalas dengan sinyal ACK di tingkat *hardware*. Jika penerima mati, sinyal ACK tidak terkirim, sehingga chip pengirim akan melakukan *retry* beberapa kali. Ketika batas percobaan habis tanpa balasan, pengirim menyimpulkan perangkat target tidak aktif dan melaporkan status gagal ke fungsi *callback* OnDataSent.

```
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')
TERIMA DARI] Node-3 (Node ID: 758106981 | Chip ID: 758106981)
Suhu      : 29.90 °C
Kelembapan : 51.00 %
=====
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.2,"kelembapan":54.8}
=====
TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu      : 1.90 °C
Kelembapan : 2.20 %
=====
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.2,"kelembapan":56.1}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.2,"kelembapan":56.1}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.3,"kelembapan":56.3}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.3,"kelembapan":56}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.2,"kelembapan":55}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.2,"kelembapan":54.9}
--> Topologi rantai mesh telah diperbarui
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.3,"kelembapan":53.7}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.3,"kelembapan":53.3}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.2,"kelembapan":52.4}
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.3,"kelembapan":52.4}
--> Topologi rantai mesh telah diperbarui
--> Koneksi Baru Terdeteksi! Node ID: 2735807929
KIRIM MESH] {"node":"Node-1","chipId":3263801740,"suhu":31.3,"kelembapan":52.6}
=====
TERIMA DARI] Node-2 (Node ID: 2735807929 | Chip ID: 2735807929)
Suhu      : 31.30 °C
Kelembapan : 51.30 %
=====
```

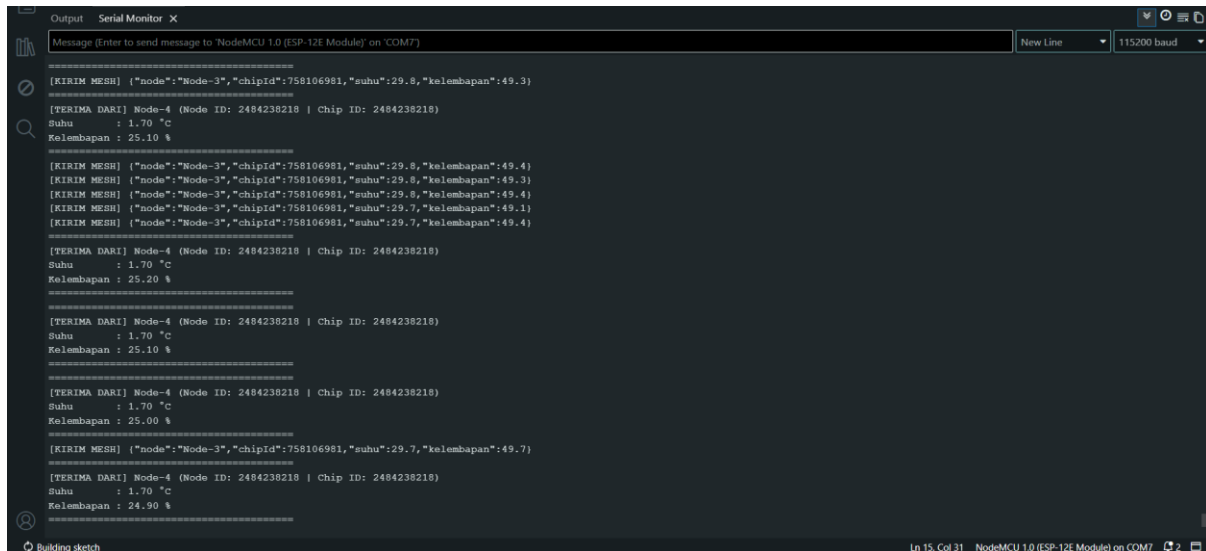
Latihan 2 (Eksperimen Dinamika Topologi & Self-Healing - Praktikum 4): Jalankan jaringan painlessMesh pada Praktikum 4 hingga seluruh board saling terhubung dan mencetak pesan --> Koneksi Baru Terdeteksi! Node ID:

- i. Tekan tombol fisik **RST** atau cabut daya salah satu board Node. Amati Serial Monitor pada board lainnya yang masih menyala. Berapa detik waktu yang dibutuhkan oleh *callback* `changedConnectionCallback()` untuk mendeteksi bahwa salah satu simpul telah keluar dari rantai mesh?

5 Detik

- ii. Sambungkan kembali daya board tersebut. Berapa lama waktu yang dibutuhkan hingga node tersebut kembali terdaftar secara otonom ke dalam topologi mesh tanpa Anda mengonfigurasi ulang jaringan? Catat hasil observasi waktu tersebut!

3 Detik



```
Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7') New Line 115200 baud

[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.8,"kelembapan":49.3)
[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.70 °C
Kelembapan : 25.10 %

[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.8,"kelembapan":49.4)
[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.8,"kelembapan":49.3)
[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.8,"kelembapan":49.4)
[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.7,"kelembapan":49.1)
[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.7,"kelembapan":49.4)

[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.70 °C
Kelembapan : 25.20 %

[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.70 °C
Kelembapan : 25.10 %

[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.70 °C
Kelembapan : 25.00 %

[KIRIM MESH] ("node":"Node-3","chipId":758106981,"suhu":29.7,"kelembapan":49.7)

[TERIMA DARI] Node-4 (Node ID: 2484238218 | Chip ID: 2484238218)
Suhu : 1.70 °C
Kelembapan : 24.90 %
```

Latihan 3 (Eksperimen Modifikasi Filter Telemetri & Ambang Batas JSON -

Praktikum 4): Pada Praktikum 4, setiap node saat ini mencetak seluruh data telemetri dari semua pengirim tanpa penyaringan.

- Modifikasilah fungsi `receivedCallback` pada salah satu board agar bertindak sebagai **Pengawas Kondisi Suhu Kritis**: program **HANYA** mencetak kartu telemetri penuh ke Serial Monitor jika suhu yang dilaporkan oleh node tetangga bernilai **di atas 32.0 °C** (lakukan pengujian dengan mendekatkan jari hangat atau meniup sensor DHT pada node pengirim).
- Jika suhu bernilai normal ($\leq 32.0\text{ °C}$), program cukup mencetak status singkat satu baris: `[NORMAL] Telemetri dari Node <chipId> aman`.
- Tuliskan blok fungsi `receivedCallback(uint32_t from, String &msg)` hasil modifikasi Anda!

5.4.3Praktikum | Arduino IDE 2.3.10

File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...

5.4.3Praktikum.ino

```
103 // Atur level log debugging (hanya tampilkan error dan status startup)
104 mesh.setDebugTypes(ERROR | STARTUP);
105
106 // Inisialisasi jaringan mesh
107 mesh.init(MESH_PREFIX, MESH_PASSWORD, &userScheduler, MESH_PORT);
108
109 // Daftarkan event callback
110 mesh.onReceive(&receivedCallback);
111 mesh.onNewConnection(&newConnectionCallback);
112 mesh.onChangedConnections(&changedConnectionCallback);
113
114 // Jadwalkan tugas pengiriman data berkala menggunakan TaskScheduler
115 userScheduler.addTask(taskSendMessage);
116 taskSendMessage.enable();
117
118 Serial.print("Mesh Node [%s] Berjalan. Menunggu pembentukan topologi...\n", nodeName);
119 }
120
121 void loop() {
122 // Wajib dipanggil rutin untuk memproses perutean paket dan sinkronisasi waktu
123 mesh.update();
124 }
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

New Line 115200 baud

[TERIMA DARI] Node-2 (Node ID: 2735807929 | Chip ID: 2735807929)
Suhu : 30.80 °C
Kelembapan : 51.80 %

[NORMAL] Telemetri dari Node 758106981 aman
[KIRIM MESH] ("node":"Node-3","chipId":"2484238218","suhu":29.3,"kelembapan":51.4)
[NORMAL] Telemetri dari Node 758106981 aman
[KIRIM MESH] ("node":"Node-3","chipId":"2484238218","suhu":29.4,"kelembapan":52)
[NORMAL] Telemetri dari Node 758106981 aman
[KIRIM MESH] ("node":"Node-3","chipId":"2484238218","suhu":29.4,"kelembapan":52)
[NORMAL] Telemetri dari Node 758106981 aman
[KIRIM MESH] ("node":"Node-3","chipId":"2484238218","suhu":29.4,"kelembapan":51.8)

Ln 124, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM5